

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination – 2024**

Course: B.Tech

Semester :IV

Branch : Artificial Intelligence & Data Science / Artificial Intelligence(AI) & Data Science / CSE (AI & DS)

Subject Code & Name: BTAIC402 Database Management System

Max Marks: 60

Date: 21/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	What is a database	CO1	1
	a. Organized collection of information that cannot be accessed, updated, and managed		
	b. Collection of data or information without organizing		
	c. Organized collection of data or information that can be accessed, updated, and managed		
	d. Organized collection of data that cannot be updated		
2	Who created the first DBMS?	CO1	1
	a. Edgar Frank Codd		
	b. Charles Bachman		
	c. Charles Babbage		
	d. Sharon B. Codd		
3	Which type of data can be stored in the database	CO1	1
	a. Image oriented data		
	b. Text, files containing data		
	c. Data in the form of audio or video		
	d. All of These		
4	Which of the following is not an example of DBMS	CO1	1
	a. MySQL		
	b. Microsoft Access		
	c. IBM DB2		
	d. Google		
5	Which of the following is not a function of the database	CO1	1
	a. Managing		
	b. Manipulating		
	c. Security for		
	d. Analysing		

	stored data	data	stored data	code		
6	What does an RDBMS consist of				CO2	1
	a. Collection of Records	b. Collection of Keys	c. Collection of Tables	d. Collection of Fields		
7	The ability to query data, as well as insert, delete, and alter tuples, is offered by				CO2	1
	a. TCL (Transaction Control Language)	b. DCL (Data Control Language)	c. DDL (Data Definition Language)	d. DML (Data Manipulation Language)		
8	Which command is used to remove a relation from an SQL				CO2	1
	a. Drop table	b. Delete	c. Purge	d. Remove		
9	For designing a normal RDBMS which of the following normal form is considered adequate				CO2	1
	a.4NF	b. 3NF	c.2NF	d.5NF		
10	Which of the following is not a NoSQL database				CO3	1
	a. SQL Server	b. MongoDB	c. Cassandra	d. None		
11	Which of the following is a wide-column store				CO3	1
	a. Cassandra	b. Riak	c. MongoDB	d. Redis		
12	Which stores are used to store information about networks, such as social connections				CO4	1
	a. Key-value	b. Wide-column	c. Document	d.Graph		
Q. 2	Solve the following.					12
A)	Explain Terms Data, Big Data and Database				CO1	6
B)	What are the benefits of Data Modeling				CO2	6
Q.3	Solve the following.					12
A)	What are the characteristics and advantages of SQL				CO2	6
B)	Explain Three Schema Architecture and Data Independence				CO2	6
Q. 4	Solve Any Two of the following.					12

A)	Explain Single Tier, Two Tier and Three Tier Architecture of DBMS	CO2	6
B)	Explain Different data types used in SQL	CO3	6
C)	What is ER Model and Explain the steps to convert ER Model into table	CO3	6
Q.5	Solve Any Two of the following.		12
A)	What are the different types of KEYS used in RDBMS? Explain with Example?	CO2	6
B)	Explain SQL Commands	CO2	6
C)	Explain 1NF,2NF,3NF and BCNF	CO3	6
Q. 6	Solve Any Two of the following.		12
A)	Explain The difference between SQL and NoSQL	CO4	6
B)	Explain the types of NoSQL databases	CO5	6
C)	Explain The CAP Theorem in distributed data systems	CO4	6
	*** End ***		